

Name: _____

Date: _____

VOLUME AND SURFACE AREA, (VOLUME OF PRISMS AND CYLINDERS)

LO: I can calculate the volume of prisms and cylinders using the formula $V = \text{area of cross-section} \times \text{length}$.

Volume of prisms and cylinders

The **volume** of a prism equals the area of its cross-section multiplied by its length. A **cylinder** is a circular prism, so its volume = $\pi r^2 h$.

Key formulae: Prism: $V = A \times l$ (A = cross-section area, l = length). Cylinder: $V = \pi r^2 h$. Volume is measured in cubic units (cm^3 , m^3).

Quick examples

●	Cuboid: $V = l \times w \times h$	rectangle cross-section $\times$ length
●	Cylinder: $V = \pi r^2 h$	circle area $\times$ height
●	Triangular prism: $V = \frac{1}{2} \times b \times h \times l$	triangle area $\times$ length
●	Volume = surface area $\times$ height	volume uses cross-section area, not surface area

Key idea

Volume of any prism = area of cross-section multiplied by the length (or height) of the prism.

$$V = \textcircled{A} \times \textcircled{l}$$

A = area of the uniform cross-section, l = length of the prism

MODELLED EXAMPLES

Study each example carefully before attempting the practice questions.

Example 1 - cuboid

Find the volume of a cuboid 8 cm by 5 cm by 3 cm.

$$\begin{aligned}\text{Cross-section area} &= 8 \times 5 = 40 \text{ cm}^2 \\ \text{Volume} &= 40 \times 3 \\ &= 120 \text{ cm}^3\end{aligned}$$

120 cm³

A cuboid is a rectangular prism.

Example 2 - cylinder

Find the volume of a cylinder with radius 4 cm and height 10 cm. Round to 1 d.p.

$$\begin{aligned}\text{Cross-section area} &= \pi r^2 = \pi \times 4^2 = 16\pi \\ \text{Volume} &= 16\pi \times 10 = 160\pi \\ &= 502.7 \text{ cm}^3\end{aligned}$$

502.7 cm³

Substitute into $V = \pi r^2 h$.

Example 3 - triangular prism

Find the volume of a triangular prism with base 6 cm, height 4 cm, and length 15 cm.

$$\begin{aligned}\text{Triangle area} &= \frac{1}{2} \times 6 \times 4 = 12 \text{ cm}^2 \\ \text{Volume} &= 12 \times 15 \\ &= 180 \text{ cm}^3\end{aligned}$$

180 cm³

Find the triangle area first, then multiply by the length.

Example 4 - trapezoidal prism

A prism has a trapezium cross-section with parallel sides 5 cm and 9 cm, height 4 cm, and length 12 cm.

$$\begin{aligned}\text{Trapezium area} &= \frac{1}{2}(5 + 9) \times 4 = 28 \text{ cm}^2 \\ \text{Volume} &= 28 \times 12 \\ &= 336 \text{ cm}^3\end{aligned}$$

336 cm³

Use the trapezium formula for the cross-section.

YOUR TURN – PRACTICE (deliberate practice)

1. Cuboids

Find the volume of each cuboid.

- a) $6 \text{ cm} \times 4 \text{ cm} \times 3 \text{ cm}$
- b) $10 \text{ cm} \times 5 \text{ cm} \times 2 \text{ cm}$
- c) $8 \text{ cm} \times 8 \text{ cm} \times 8 \text{ cm}$
- d) $12 \text{ cm} \times 3 \text{ cm} \times 5 \text{ cm}$

2. Cylinders

Find the volume of each cylinder. Round to 1 d.p.

- a) $r = 3 \text{ cm}$, $h = 10 \text{ cm}$
- b) $r = 5 \text{ cm}$, $h = 8 \text{ cm}$
- c) $r = 2 \text{ cm}$, $h = 15 \text{ cm}$
- d) $r = 7 \text{ cm}$, $h = 6 \text{ cm}$

3. Triangular prisms

Find the volume of each triangular prism.

- a) Base 8, height 5, length 12 cm
- b) Base 10, height 6, length 9 cm
- c) Base 7, height 4, length 20 cm
- d) Base 12, height 3, length 8 cm

4. Other prisms

Find the volume using $V = A \times l$.

- a) Trapezium: parallel sides 4, 8; height 3; length 10 cm
- b) Cross-section area 25 cm^2 , length 14 cm
- c) L-shape cross-section area 32 cm^2 , length 6 cm
- d) Hexagonal cross-section area 42 cm^2 , length 9 cm

5. Mixed volume problems

Solve each problem.

COMMON MISCONCEPTIONS – SPOT THE MISTAKE

Each statement shows a student's answer. Tick () the ones that are correct. If it is wrong, write the correct answer.

a) Volume of a prism = area of cross-section $\times$ length ☐

Correct answer: _____

Reason: _____

b) Volume of a cylinder = $2\pi rh$ ☐

Correct answer: _____

Reason: _____

c) $V = \pi r^2 h$ requires the radius, not the diameter ☐

Correct answer: _____

Reason: _____

d) Volume and surface area use the same formula ☐

Correct answer: _____

Reason: _____

e) Volume is measured in cubic units such as cm^3 ☐

Correct answer: _____

Reason: _____

6. CHALLENGE – WORD PROBLEMS (apply your skills)

a) A cylindrical tank of radius **0.5 m** and height **1.2 m** is filled with water. The water is poured into a rectangular tank measuring **1 m by 0.8 m**. What depth does the water reach? Give your answer to the nearest cm.

Expression: _____

Simplified: _____

b) A triangular prism-shaped chocolate bar has a cross-section that is an equilateral triangle with side **4 cm**. The bar is **18 cm** long. If chocolate weighs **1.3 g per cm^3** , find the mass of the bar to the nearest gram.

Expression: _____

Simplified: _____

TOP TIPS

Read the question carefully. Show all your working step by step.
Check your answer makes sense.

CHECK YOUR WORK

Have I shown all my working clearly?

Did I check my answer using a different method?

Does my answer look reasonable?

